TAKE-HOME ASSIGNMENT

Dyson Data Analytics & Insights Engineer

Take-home assignment

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Executive Summary

KEY FINDINGS

~6x

Filter Wash reminders flag, and come before, airway blockages

more likely in a clean that shows the reminder

27%

The SV53 rarely warns before a blockage error

of SV53 blockage errors were warned first (SV50: 92%)

3–5x

Returned SV53s log more blockages, led by sub-model SV53-AB

more blockages than normal SV53s

74%

The SV50 often sends no battery temperature

of SV50 cleans (SV53: 3%); it also sends no dust or motor data

MOST IMPORTANT INSIGHT

The pattern points to restricted airflow as the central story. The filter reminder is a cheap early signal of it, and the SV53 often skips its own warning.

BUSINESS IMPLICATIONS

- Treat the filter reminder as an early warning, not a nuisance alert
- Report fault rates per sub-model before judging reliability

ENGINEERING IMPLICATIONS

- Review why the SV53 skips the blockage warning
- Make every model send battery temperature, dust and motor data

Returned = the Return/Repair group (all model SV53). 232 machines (186 normal, 46 returned).

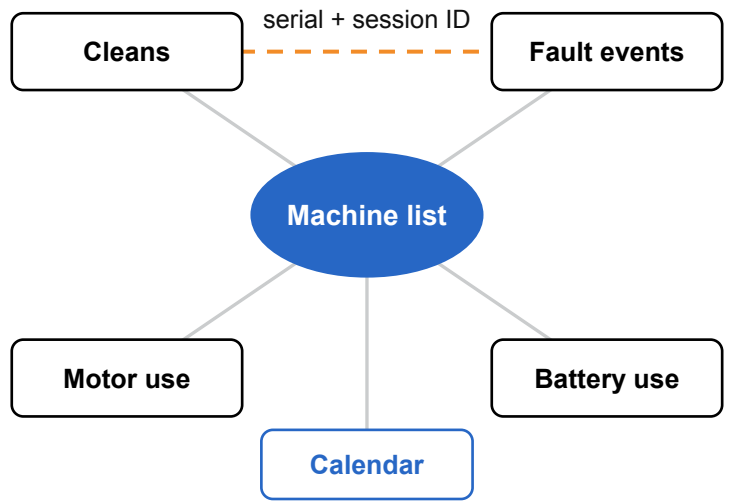
Source: reports/insights/key_numbers.csv (f_ rows); deck_numbers.md

Data Understanding and Modelling

232 machines (186 normal, 46 returned or repaired)

Table	One row is	Built from	Used for
Machine list `dim_machine`	One machine: model, region, group	Serials in all four files	Grouping and filtering every result
Calendar `dim_date`	One day	Generated	Trends by week or month
Cleans `fact_session`	One cleaning session	Cleaning sessions file	Rates per 100 cleans; battery and dust readings
Fault events `fact_fault`	One fault, after removing copies	Faults file	Blockage and reminder counts; event order
Motor use `fact_motor_period`	Change between two motor snapshots	Motor file (lifetime counters)	Motor minutes per day
Battery use `fact_battery_period`	Change between two battery snapshots	Battery file (lifetime counters)	Battery heat exposure

STAR SCHEMA



JOIN STRATEGY

1. **Machine serial:** links every table to the machine list; normal vs returned is a column, so one code path serves both
2. **Event date:** links every event to the calendar
3. **Serial + session ID:** links each fault to the clean it happened in (upload dates within 24 h; 99.4% matched)

IN PRODUCTION (LAKEHOUSE)

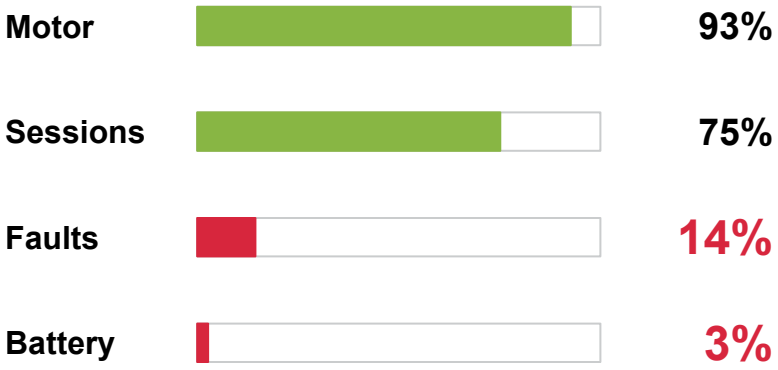
Ingest uploads land daily › **Bronze** raw files, never edited › **Silver** parse, de-duplicate, flag, counters to deltas › **Gold** the six tables above › **Visualise** Tableau; DQ checks gate each step

Assumptions to combine the data: no fault rows = zero faults; identical rows = re-uploads; a lifetime counter that drops = bad period, not a reset.

Source: repo src/ (pipeline), docs/erd, docs/assumptions_log.md

Data Quality Assessment

SHARE OF RAW ROWS USABLE



How the 3% of battery data is used
958 usable rows from 70 machines; 57 of them (15 SV50, 42 SV53) have a full period. Used only for the battery-heat check in Finding 4, rated low confidence. Finding 4's main result uses the 35,473 cleans in the sessions table.

Duplicates

Three kinds: exact copies (every column identical, same timestamp to 1/100 s) are 78% of fault rows; re-uploads (same event, later upload) 5%; session join fan-out 25% of session rows.

Action: **Kept one per event**

Missing data

95% of raw battery rows have no data payload, all sent by 14 machines re-sending empty records. Firmware version is blank in 99.5% of cleans.

Action: **Excluded and flagged**

Invalid values

Rule-based: dust value 65535 (the largest 16-bit number, a usual "no reading" code; inferred) in 11% of cleans; 194 zero-length cleans; 3 battery % outside 0–100.

Action: **Set to missing, row kept**

Outliers

Machines beyond 3× the interquartile range: 10 on cleans per week and 10 on fault rate.

Action: **Flagged and kept; leave-one-out checks**

Limitations

No return reasons or dates. The SV50 sends no dust, battery-temperature or motor data. Several field meanings are not stated.

Action: **Compare within one model**

Fix at source

De-duplicate at ingestion, reject empty payloads, require the same fields from every model, document codes and units.

Action: **Quickest win**

Biggest fix at the source: de-duplicate uploads, reject empty battery records, and make every model send the same fields.

Nothing was deleted: problem rows are flagged, so every number traces back to the raw files. Rules are in the repo (dq_summary.csv).

Source: key_numbers.csv (dq_ rows); reports/dq/dq_summary.csv

Analytical Insights

INSIGHT	REASON	IMPLICATION	HOW IT HELPS
1 · FAULT CO-OCCURRENCE **Filter Wash reminders flag, and come before, airway blockages** **~6x** as likely to also log an airway blockage	Compared each machine's cleans with and without the reminder: 11.8% vs 0.9% logged a blockage. About 3x in the next 7 days.	Both alerts likely react to the same restricted airflow (untested).	Use the reminder as an early warning; test firmer prompts and measure blockages after.
2 · FAULT SEVERITY BY MODEL **The SV53 rarely warns before a blockage error** **27%** of SV53 blockage errors were warned first (SV50: 92%)	Put each clean's faults in time order and checked whether a blockage error had a "still running" warning first.	The SV53 may detect blockage differently, or skip the warning stage (untested).	Firmware review of SV53 detection. The error follows the warning in ~4s, so protection must be automatic.
3 · FAULT FREQUENCY **Returned SV53s log more blockages, led by submodel SV53-AB** **3–5x** vs normal SV53s	Blockage rate per sub-model, then matched the sub-model mix: 4.7x falls to 3.1x.	Returned machines are mostly SV53-AB (58% vs 31%), which blocks most: 8.2 vs 0.6–2.3 per 100 cleans.	Focus airflow review on SV53-AB; report fault rates per sub-model.
4 · BATTERY BEHAVIOUR **The SV50 often sends no battery temperature** **74%** of SV50 cleans	Checked which fields each model fills in; returned vs normal differ by only 0.5 points, so it's the model.	The SV50 doesn't send temperature, dust or motor data, yet 9 of 15 SV50 batteries went above 45°C (SV53: 5 of 42).	Require the same telemetry from every model, with completeness checks on arrival (a lead to check).

Additional observations: Returned machines ran ~2x the motor minutes per day. Dust readings and motor pressure or temperature did not explain blockages.

Like with like: returned vs normal uses SV53 only; models are compared on normal machines. (untested) = an interpretation the data can't confirm.

Source: key_numbers.csv (f_ rows); deck_numbers.md; insight_cards.md (final insights)

Recommendations

NEXT ANALYSES

1	Match blockages and filter reminders to repair reasons	Insight 1, 3
2	Test firmer filter reminders; measure blockages after	Insight 1
3	Confirm with firmware whether the SV53 skips the warning	Insight 2
4	Investigate the SV53-AB airflow path	Insight 3

DATA IMPROVEMENTS

1	Record return reasons and return dates
2	Make the SV50 send battery temperature, dust and motor data
3	Stop duplicate fault uploads and empty battery records
4	Send firmware version; document what each field means

FUTURE OPPORTUNITIES

Early warning
Predict blockages from reminder and warning events

Battery health
Monitor heat on every model once the SV50 reports temperature

Sub-model tracking
Report fault rates per sub-model, not just per model

Dashboard: a Tableau insights dashboard and data-quality explorer (screenshots in the repo).

Source: reports/insights/insight_cards.md (final insights); docs/assumptions_log.md